

The empirical recurrence for OEIS A237960, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A237960 records “Empirical recurrence of order 82” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1422$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 84$. Its last coefficient is $c_{82} = -1905412513202176 \neq 0$, so no further tail satisfies anything shorter; any order-82 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A237960 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the lower median minus the minimum of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1422, 10942, 81483, 645156, 5242451, 41361664, 324579268, 2567231540, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:39:03, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1422$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1422$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 82: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 82.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2844$ exact terms from $n = 2$ on, it returns order 82 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	0	c_{22}	89048330242322	c_{43}	0	c_{64}	−232048240596696914
c_2	0	c_{23}	0	c_{44}	−76460726588292326524	c_{65}	0
c_3	0	c_{24}	−541816238292384	c_{45}	0	c_{66}	−326603404925588705
c_4	3740	c_{25}	0	c_{46}	43205151844051503314	c_{67}	0
c_5	0	c_{26}	−3656290194080835	c_{47}	0	c_{68}	−217065743980989952
c_6	59736	c_{27}	0	c_{48}	133131977207787311756	c_{69}	0
c_7	0	c_{28}	15721553934787578	c_{49}	0	c_{70}	−52680072751015231
c_8	−2560066	c_{29}	0	c_{50}	112064495935731672476	c_{71}	0
c_9	0	c_{30}	98696328062585090	c_{51}	0	c_{72}	27442499770463027
c_{10}	−43466447	c_{31}	0	c_{52}	36909332708860727336	c_{73}	0
c_{11}	0	c_{32}	−255506281484315386	c_{53}	0	c_{74}	−7617145698949529
c_{12}	772997615	c_{33}	0	c_{54}	−89484551886316197760	c_{75}	0
c_{13}	0	c_{34}	−1715992049731996804	c_{55}	0	c_{76}	−4755553304560271
c_{14}	11380678545	c_{35}	0	c_{56}	−85333287930431067744	c_{77}	0
c_{15}	0	c_{36}	1637270877824323414	c_{57}	0	c_{78}	−1703500877502873
c_{16}	−128503967290	c_{37}	0	c_{58}	282855056904828405872	c_{79}	0
c_{17}	0	c_{38}	16324034030014699635	c_{59}	0	c_{80}	−1437793438295654
c_{18}	−1342418039875	c_{39}	0	c_{60}	557124431874091277760	c_{81}	0
c_{19}	0	c_{40}	5174215296562511768	c_{61}	0	c_{82}	−190541251320217
c_{20}	11110233453890	c_{41}	0	c_{62}	204182297177913434112		
c_{21}	0	c_{42}	−64058453614690150432	c_{63}	0		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 84$, and it is the recurrence OEIS A237960 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{82} - c_1x^{82-1} - \dots - c_{82}$. Its constant term is $-c_{82} = 1905412513202176$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 82 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 82, so $\deg q = 82$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 82, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 82, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1905412513202176 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-82 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237960.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.